

Q11 Implement product cipher transposition operation.

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In [1]: def caesar(text, shift):
        result = ""
        for ch in text:
            if ch.isalpha():
                result += chr((ord(ch.upper()) - ord('A') + shift) % 26 + ord('A'))
        return result

    def caesar_decrypt(text, shift):
        result = ""
        for ch in text:
            if ch.isalpha():
                result += chr((ord(ch.upper()) - ord('A') - shift) % 26 + ord('A'))
        return result

    def rail_fence(text, key):
        rail = ['' for _ in range(key)]
        row, down = 0, True
        for ch in text:
            rail[row] += ch
            if row == 0: down = True
            elif row == key - 1: down = False
            row += 1 if down else -1
        return ''.join(rail)

    def rail_fence_decrypt(cipher, key):
        idx_rail = [[] for _ in range(key)]
        row, down = 0, True
        for i in range(len(cipher)):
            idx_rail[row].append(i)
            if row == 0: down = True
            elif row == key - 1: down = False
            row += 1 if down else -1

        result = [''] * len(cipher)
        current_char_idx = 0
        for r in range(key):
            for pos in idx_rail[r]:
                result[pos] = cipher[current_char_idx]
                current_char_idx += 1
        return ''.join(result)

    def encryption(text, shift, rails):
        step1 = caesar(text, shift)
        step2 = rail_fence(step1, rails)
        return step2

    def decryption(cipher, shift, rails):
        step1 = rail_fence_decrypt(cipher, rails)
        step2 = caesar_decrypt(step1, shift)
        return step2
```

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In [2]: text = input("Enter plaintext: ")
        shift = int(input("Enter shift: "))
        rails = int(input("Enter rails: "))

        encrypted = encryption(text, shift, rails)
        print("Encrypted:", encrypted)

        decrypted = decryption(encrypted, shift, rails)
        print("Decrypted:", decrypted)
```

Encrypted: JQFGNYTNQ
Decrypted: HELLOWORD